MS4092 COMPREHENSIVE EXAMINATION

General Engineering

Semester II 2022/2023

Saturday, 4th Mar 2023, Duration: 32 minutes

Closed Book and Closed Note

Answer all the questions on this problem sheet. State your answer by circling the letter in front of the most correct/accurate answer. Correct answers get additional scores, while wrong answers or questions that are not answered get zero score. If necessary, you may use the back page of the question sheet as 'scratch paper'. Don't forget to fill in your name, student identity number, and/or your signature at the top of EVERY page (in the space provided).

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1. Evaluate the following limit:

$$\lim_{x\to 5}\left(\frac{x^2-25}{x-5}\right) = \lim_{x\to 5}\frac{(x-5)(x+5)}{(x-5)} = 10$$

a) 0
b) 5
c) 10
d) ∞

2. If V is a velocity vector $(u\hat{\imath} + v\hat{\jmath} + w\hat{k})$, and T is temperature (scalar), which equation has the same meaning with expression $u\frac{\partial T}{\partial x} + v\frac{\partial T}{\partial y} + w\frac{\partial T}{\partial z}$

a) $\frac{DT}{DT}$
b) $(V\cdot\nabla)T$
c) $T\,(\nabla V)$
d) $(T\cdot\nabla)V$

3. Among the following quantities, which one is not being a vector quantity?

a) Force
b) Velocity
c) Energy
d) Stress

4. It is desired to choose a resistance temperature detector (RTD) transducer for a measurement. The actual value of the temperature is not important because the initial temperature is known by other means, but the change in temperature as the test item is heated is important. Which statement is true about the selection of the RTD?

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a) The precision is important, but accuracy is not important
b) The accuracy is important, but the precision is not important
c) Neither the accuracy nor the precision is important
d) Both accuracy and precision are important

5. Determine the following indefinite integral:

$$\int \frac{x^2 + x + 4}{x^2} dx$$

a) $\frac{x}{4} + \ln|x| - \frac{4}{x} + C$
b) $\frac{-x}{2} + \log x - 8x + C$
c) $\frac{x^2}{2} + \ln|x| - \frac{2}{x^2} + C$
d) $\frac{x^2}{2} + \ln|x| - \frac{4}{x} + C$

$\int 1 + \frac{1}{x} + \frac{4}{x^2} dx$

$= x + \ln|x| - \frac{4}{x} + C$

6. Using Newton's method with an initial estimate of $x_0 = 2$ through two iterations, what is the root of the following equation?

$$f = x^2 - 4x - 5$$

a) 2.47
b) 2.60
c) 2.63
d) 16.7

$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$

$f' = 2x - 4$

kalau $x_0 = 2 \rightarrow f'(x_0) = 0 \rightarrow$ hasil infinit

$x_0 = 2.01$

$f = -8,999$

$f' = 0.02$

$x_1 = 2.01 - \frac{(-8,999)}{0.02}$

$x_1 = 451,96$

$x_1 = 451,96$

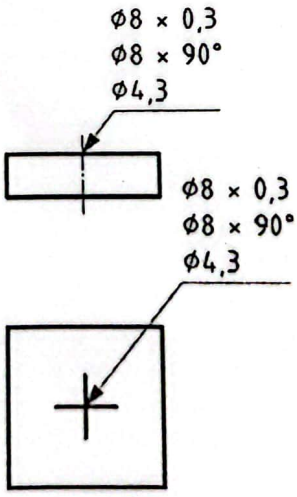
$f = 202465$

$f' = 899,92$

$x_2 = 451.96 - \frac{202465}{899,92}$

$x_2 = 226,98$

7. On the following simplified drawing, the pointed profile is...



a) Counterbore hole → ada penunjuk dalamnya bore berapa
b) Countersink hole → ada penunjuk sudut
c) Through hole → ada ket. thru
d) None of the other answers is correct.

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8. In the following iterative program, what is the final value of variable k? (medium)

```
a = 3; b = 1
x = a
k = 0
l = 0

while (l<x):
        k = k*x*l + x^l
        l = l + b

print (m)
```

$l=0$
$k=1 \rightarrow k=0.3.0+3^0$
$l=1$
$k=6 \rightarrow k=1.3.1+3^1$
$l=2$
$k=45 \rightarrow k=6.3.2+3^2$

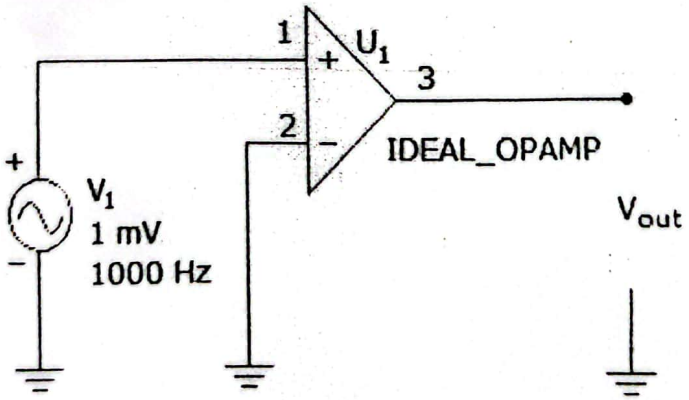
a) 1
b) 6
c) 27
d) 45

9. A curve construct from $y = 2x-x^2$ function will get the maximum value at point?

a) (1,1)
b) (1,2)
c) (2,2)
d) (2,1)

$\frac{dy}{dx}=0$
$2-2x=0$
$x=1$
$y=2.1-1^2=1$

10. What is output waveform of the circuit?



a) Sine wave
b) Square Wave
c) Constant Voltage
d) Triangle Wave

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11. Flat plane, α, is normal to vector n = (3, 5, 2), coordinates of point A and B are (1, 2, 3) and (7, 6, 5), respectively. Find the length of orthogonal projection $\overline{AB}$ on the α plane

a) 3.09

b) 6.81

c) 7.48

d) None of the above answer is correct

$\vec{AB} = \langle 6,4,2 \rangle$

$|\vec{AB}| = \sqrt{56}$

$|\vec{n}| = \sqrt{38}$

$\vec{AB} \cdot \vec{n} = |\vec{AB}|\,|\vec{n}| \cos\theta$

$\cos\theta = \frac{6.3+4.5+2.2}{\sqrt{56}\cdot\sqrt{38}} = 0,91$

$\sin\theta = 0,41$

AB projection $= |\vec{AB}| \sin\theta = 3,09$

12. A particle moves in the *x*-*y* plane. After *t* seconds, the x and y coordinates of the particle's location are in x = 8 sin (*t*) and y = 6 cos (*t*). Which of the following equations describes the path of particle?

a) $36x^2 + 64y^2 = 2304$

b) $36x^2 - 64y^2 = 2304$

c) $64x^2 + 36y^2 = 2304$

d) $64x^2 - 36y^2 = 2304$

$\sin t = \frac{x}{8}$

$\cos t = \frac{y}{6}$

$\sin^2 t + \cos^2 t = 1$

$\frac{x^2}{64} + \frac{y^2}{36} = 1$

$36x^2 + 64y^2 = 2304$

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